Yingru Li

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CURRICULUM VITAE

EDUCATION

The Chinese University of Hong Kong (CUHK) 2025
Ph.D. in Computer and Information Engineering. Advisor: Zhi-Quan (Tom) Luo.
Committee: Xinyun Chen, Baoxiang Wang, John C.S. Lui, Benjamin Van Roy (Stanford & DeepMind).

Huazhong University of Science and Technology (HUST) 2020
B.E./M.E. in Computer Science. Ranked 1st/134 overall; 1st/26 in Software and Theoretical CS.

PROFESSIONAL EXPERIENCE

xAI (now SpaceX Super Intelligence) Singapore
Member of Technical Staff Dec 2025 – Present

- Core contributor to RL fundamentals for large-scale ("big-run") reinforcement learning of the Grok models.
  - *Science of RL training* — core algorithms and recipes; RL scaling from first principles; optimizer and numerics at scale.
  - *Stability and efficiency* — identified and resolved critical training instabilities affecting every stage of the RL pipeline, contributing to one of the longest stable RL runs to date.
  - *Long horizon* — training recipes for long-horizon agentic RL.
  - *On-policy distillation* — training recipes and supporting infrastructure.
  - *Infrastructure* — built and scaled these recipes in large-scale distributed RL systems.

ByteDance Singapore
LLM Algorithm Research Scientist 2025 – Dec 2025

- Led research on RL training stability and efficiency for LLM reasoning and agents.
  - Diagnosed and formalized RL training collapse, tracing it to implicit off-policyness from the training–inference mismatch; developed Trust Region Masking (ICML'26), Optimal Token Baseline (ICML'26), Dynamic Vocabulary Pruning, and a simple LR-scheduling fix.
  - Core contributor to veRL (20k+ GitHub stars): authored the Rollout Correction framework and the Optimal Token Baseline estimator (see Research Impact).
  - Co-led end-to-end RL for multi-turn tool-integrated reasoning (SimpleTIR, ICLR'26) and a Triton kernel-generation agent (Dr. Kernel, ICML'26).

Tencent AI & Robotics X Shenzhen, China
Student Research Staff Member, Agent Center 2019 – 2022

- Led research on scalable exploration in deep RL—HyperAgent (ICML'24), HyperDQN (ICLR'22)—and stable off-policy learning—Divergence-Augmented Policy Optimization (NeurIPS'19).

Microsoft Research Beijing, China
Research Intern, Theory Center 2016

- Combinatorial optimization and reinforcement / bandit learning algorithms.

RESEARCH IMPACT & ADOPTION

The RL-collapse / training–inference mismatch line of work (*When Speed Kills Stability*) and its fix—masked importance sampling (MIS), made principled as Trust Region Masking (TRM)—have been adopted across open-source RL stacks and frontier models:

- **DeepSeek-V3.2** adopts geometric sequence masking to address the training–inference mismatch and off-policy instability.
- **veRL** (v0.6.0+) — authored the Rollout Correction framework for general off-policy training, which corrects the gap between the rollout policy and the training policy (training–inference mismatch, stale asynchronous rollouts, replayed data) via bounded importance weighting and outlier masking at token, sequence, or geometric level, and serves as the base on which veRL's newer off-policy algorithms (e.g., score centering and IcePop) are built; also authored the Optimal Token Baseline, a variance-reducing token-level baseline; Geo-MIS / Geo-RS and Trust Region Masking are first-class algorithms.
- Integrated into Slime and Miles; discussed in a vLLM engineering blog and an SGLang blog.
- Used in Cognition's SWE-grep and discussed in the Qwen team technical report.
- Cited and built upon by follow-up work, e.g., *Defeating the Training-Inference Mismatch via FP16* (arXiv:2510.26788).
- Widely shared and discussed in the research community: original thread on X; a community quote-post citing our finding to explain reported A100 RL instability reached 84K views.

PUBLICATIONS

Full list on Google Scholar. *equal contribution; †corresponding author.

Peer-Reviewed Conference & Journal Publications

1. **The Optimal Token Baseline: Variance Reduction for Long-Horizon LLM-RL**
   Yingru Li, Jiawei Xu, Ziniu Li, Jiacai Liu, Wei Liu, Yuxuan Tong, Longtao Zheng, Zhenghai Xue, Yaxiang Zhang, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *ICML 2026*
2. **Trust Region Masking for Long-Horizon LLM Reinforcement Learning**
   Yingru Li, Jiacai Liu, Jiawei Xu, Yuxuan Tong, Ziniu Li, Qian Liu, Baoxiang Wang. *ICML 2026*
3. **Optimistic Thompson Sampling for No-Regret Learning in Unknown Games**
   Yingru Li, Liangqi Liu, Wenqiang Pu, Hao Liang, Zhi-Quan Luo. *IEEE Transactions on Signal Processing, 2026*
4. **On the Scaling Flaws of Verifier-Guided Beam Search in Mathematical Reasoning**
   Fei Yu, **Yingru Li**, Benyou Wang. *Transactions on Machine Learning Research (TMLR), 2026*
5. **Dr. Kernel: Reinforcement Learning Done Right for Triton Kernel Generations**
   Wei Liu, Jiawei Xu, **Yingru Li**, Longtao Zheng, Tianjian Li, Qian Liu, Junxian He. *ICML 2026*
6. **Harnessing Uncertainty: Entropy-Modulated Policy Gradients for Long-Horizon LLM Agents**
   Jiawei Wang, Jiacai Liu, Yuqian Fu, **Yingru Li**, Xintao Wang, Yuan Lin, Yu Yue, Lin Zhang, Yang Wang, Ke Wang. *ICML 2026*
7. **SimpleTIR: End-to-End Reinforcement Learning for Multi-Turn Tool-Integrated Reasoning**
   Zhenghai Xue, Longtao Zheng, Qian Liu, **Yingru Li**, Xiaosen Zheng, Zejun Ma, Bo An. *ICLR 2026*
8. **Scalable Exploration via Ensemble++**
   Yingru Li, Jiawei Xu, Baoxiang Wang, Zhi-Quan Luo. *NeurIPS 2025*
9. **Owl: Optimized Workforce Learning for General Multi-Agent Assistance in Real-World Task Automation**
   Mengkang Hu, Yuhang Zhou, Wendong Fan, Yuzhou Nie, Bowei Xia, Tao Sun, Ziyu Ye, Zhaoxuan Jin, **Yingru Li**, Qiguang Chen, Zeyu Zhang, Yifeng Wang, Qianshuo Ye, Bernard Ghanem, Ping Luo, Guohao Li. *NeurIPS 2025*
10. **Q-Star Meets Scalable Posterior Sampling: Bridging Theory and Practice via HyperAgent**
    Yingru Li, Jiawei Xu, Lei Han, Zhi-Quan Luo. *ICML 2024*
11. **Prior-Dependent Analysis of Posterior Sampling RL with Function Approximation**
    Yingru Li, Zhi-Quan Luo. *AISTATS 2024*
12. **HyperDQN: A Randomized Exploration Method for Deep Reinforcement Learning**
    Ziniu Li, **Yingru Li**†, Yushun Zhang, Tong Zhang, Zhi-Quan Luo. *ICLR 2022*

13. **Divergence-Augmented Policy Optimization**
Qing Wang*, **Yingru Li***, Jiechao Xiong, Tong Zhang. *NeurIPS 2019*
14. **Hidden Community Detection in Social Networks**
Kun He, **Yingru Li**, Sucheta Soundarajan, John E. Hopcroft. *Information Sciences, vol. 425, pp. 92–106, 2018*

Preprints & Technical Reports

1. **When Speed Kills Stability: Demystifying RL Collapse from the Training-Inference Mismatch**
Jiacai Liu*, **Yingru Li***,†, Yuqian Fu, Jiawei Wang, Qian Liu, Yu Shen. *Technical Blog, 2025*
2. **Beyond Precision: Training-Inference Mismatch is an Optimization Problem and Simple LR Scheduling Fixes It**
Yaxiang Zhang*, **Yingru Li***,†, Jiacai Liu, Jiawei Xu, Ziniu Li, Qian Liu, Haoyuan Li. *arXiv 2026*
3. **Learning an Opponent-Aware Anti-Jamming Strategy via Online Convex Optimization**
Liangqi Liu, Wenqiang Pu, **Yingru Li**, Zhi-Quan Luo. *Under review (minor revision), IEEE Transactions on Signal Processing; arXiv 2026*
4. **Dynamic Vocabulary Pruning: Stable LLM-RL by Taming the Tail**
Yingru Li, Jiawei Xu, Jiacai Liu, Yuxuan Tong, Ziniu Li, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *arXiv 2025*
5. **A Note on Hybrid Online RL and Imitation Learning for LLMs: Formulations and Algorithms**
Yingru Li, Ziniu Li, Jiacai Liu. *arXiv 2025*
6. **Logit Dynamics in Softmax Policy Gradient Methods**
Yingru Li. *arXiv 2025*
7. **Uncertainty-Aware Search and Value Models: Mitigating Search Scaling Flaws in LLMs**
Fei Yu, **Yingru Li**, Benyou Wang. *arXiv 2025*
8. **Probability Tools for Sequential Random Projection**
Yingru Li. *arXiv 2024*
9. **Simple, Unified Analysis of Johnson-Lindenstrauss with Applications**
Yingru Li. *arXiv 2024*

Selected Writing

A theory of RL training: why post-training updates are sparse and low-rank (a covariance capped by the Fisher, confined to the directions that separate the tokens competing at forks) and how asynchronous RL should scale (a step size that follows the optimizer and a queue sized in rollouts). Posts at richardli.xyz/post.

- *The Shape of a Post-Training Update*, four parts: The Gradient Is a Covariance; Sparsity Requires Score Centering; What Policy Gradient Can Reach; Sparse in the Vocabulary, Low-Rank in Hidden Space (2026).
- *Off-Policy RL at Scale*, two parts: Around the Sampler: Tangent, Remainder and KL; Keeping the Off-Policy Gap Fixed as RL Scales (2026).
- *Information Bandwidth in Reinforcement Learning* (2025).

Patents

- Liangqi Liu, Wenqiang Pu, **Yingru Li**, Zhi-Quan Luo. *Game-Strategy Update Method, Device, and Medium for Unknown Matrix Games.* Chinese invention patent application CN117787419A (Shenzhen Research Institute of Big Data; published Mar. 2024).

Awards & Honors

- ICML 2026 Gold Reviewer.
- Best Paper Award, 3rd Doctoral and Postdoctoral Daoyuan Academic Forum, 2024.
- Best Student Paper, IEEE Sensor Array and Multichannel Signal Processing Workshop (SAM), 2024.

- SRIBD Ph.D. Fellowship (Gold Class), Shenzhen Research Institute of Big Data, 2023.
- Presidential Ph.D. Fellowship, The Chinese University of Hong Kong, 2019–2023.
- Tencent AI Ph.D. Fellowship, Tencent & The Chinese University of Hong Kong, 2018.
- National Scholarship (China), top 0.2% nationwide, 2018.
- Qiming Star Award — 1 of 5 recipients from 7,112 undergraduates, HUST, 2016.
- First Prize, Parallel Computation and Application Contest (PAC), Intel & CCF, 2015.
- First Prize, China National Mathematics Olympiad (Province-level Math League), 2012.

Invited Talks

- *Revisiting Trust Region Policy Optimization in the Era of LLMs* — QingkeAI, Dec. 2025.
- *LLM-RL Instability and the Training–Inference Mismatch* — SLAI and the Bay Area RL community, Nov. 2025.
- *Tackling Data Scarcity for Trustworthy Agents* — ETH Zurich, Nov. 2024.
- *Exploration at Scale: Theory, Algorithms & Applications* — MIT (Jul. 2024), Princeton University (May 2024), INFORMS Annual Meeting (Seattle, Oct. 2024), ISMP (Montréal, Jul. 2024), INFORMS Optimization Society Conference, Rice University (Mar. 2024), RLChina (Jun. 2024); contributed talk, 3rd Daoyuan Academic Forum (Jan. 2024).
- *No-Regret Learning in Unknown Games with Applications* — RL Theory Student Workshop, Nanjing University (Aug. 2022); contributed talk, 2nd Daoyuan Academic Forum (Aug. 2022).
- *HyperDQN: A Randomized Exploration Method for Deep RL* — contributed talk, NeurIPS Workshop on Ecological Theory of RL, Dec. 2021.

Professional Service

Reviewer (record on OpenReview, ~Yingru_Li1):

Venue	Years	Papers
NeurIPS	2023–2026	22
ICLR	2024–2026	10
ICML	2025–2026	11
AISTATS	2025	2
TMLR	ongoing	1
Total		46

- AISTATS 2026 reviewer pool.
- 28 additional papers across 9 workshops at ICLR, ICML, and NeurIPS.
- Session Chair, 2 sessions on "Integrating Generative AI with Sequential Decision-making," INFORMS Annual Meeting 2024.
- Organizer, Reinforcement Learning Seminar, CUHK (8 semesters, 2019–2022).

Teaching

Teaching Assistant, The Chinese University of Hong Kong (Shenzhen): Stochastic Processes (Fall 2018), Optimization II (Spring 2019), Distributed and Parallel Computing (Fall 2019), Reinforcement Learning (Fall 2020), Matrix Analysis (Spring 2021), Deep Learning and Their Applications (Spring 2022).